

## Research Interests

*Robotics:* Grasping & Dexterous Manipulation, Soft Robotics, Tactile Sensing, Motion Planning  
*Graphics:* Animation, AR / VR, Discrete Differential Geometry, Physics-Based Simulation & Modeling  
*Optimization:* Quadratic Programming, Non-Convex Methods  
*Artificial Intelligence:* Reinforcement Learning, Deep Learning, Heuristic Search, Statistical Inference  
*Imaging & Vision:* 3D Vision, Segmentation & Reconstruction, Stereo Photography

## Education

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|---------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------|
| <b>Carnegie Mellon University</b><br>Ph.D. Computer Science<br>Advisor: Nancy Pollard                               | <b>GPA: N/A</b>   | <b>Graduation: 2026</b> |
| <b>University of California, Los Angeles</b><br>M.S. Computer Science<br>Advisor: Demetri Terzopoulos               | <b>GPA: 3.814</b> | <b>Graduation: 2018</b> |
| <b>University of California, Los Angeles</b><br>B.S. Computer Science & Engineering<br>B.S. Mathematics & Economics | <b>GPA: 3.668</b> | <b>Graduation: 2016</b> |

## Research Appointments

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|-------------------------------------------------------------------|--------------------------------------|
| Visiting Researcher   Meta AI / RL Research                       | <i>May 2022 — Present</i>            |
| Doctoral Research Assistant   CMU Foam Robotics Lab               | <i>September 2020 — Present</i>      |
| Research Scientist   NeuralX Incorporated                         | <i>March 2019 — August 2020</i>      |
| Staff Research Associate   UCLA Computer Graphics & Vision Lab    | <i>January 2019</i>                  |
| Graduate Student Researcher   UCLA Computer Graphics & Vision Lab | <i>November 2017 — December 2018</i> |

## Publications (Reverse Chronological)

### Contact Transfer: A Direct, User-Driven Method for Human to Robot Transfer of Grasps and Manipulations

**Arjun Lakshmipathy**, Dominik Bauer, Cornelia Bauer, Nancy S. Pollard  
*Proc. of IEEE International Conference on Robotics and Automation (ICRA), Philadelphia, USA, May 2022*

### Towards Very Low-Cost Iterative Prototyping for Fully Printable Dexterous Soft Robotic Hands

Dominik Bauer, Cornelia Bauer, **Arjun Lakshmipathy**, Roberto Shu, Nancy S. Pollard  
*Proc. of IEEE International Conference on Soft Robotics (ROBOSOFT), Edinburgh, UK, April 2022*

### Fully Printable Low-Cost Dexterous Soft Robotic Manipulators for Agriculture

Dominik Bauer, Cornelia Bauer, **Arjun Lakshmipathy**, Nancy S. Pollard  
*AAAI Conference on Artificial Intelligence - AI for Agriculture and Food Systems Workshop (AAAI AIFS), Vancouver, Canada, February 2022*

### Contact Tracing: A Low Cost Reconstruction Framework for Surface Contact Interpolation

**Arjun Lakshmipathy**, Dominik Bauer, Nancy S. Pollard  
*Proc. of IEEE International Conference on Intelligent Robots and Systems (IROS), Prague, Czech Republic, September 2021*

### Locally-Connected, Irregular Deep Neural Networks for Biomimetic Active Vision in a Simulated Human

Masaki Nakada, Honglin Chen, **Arjun Lakshmipathy**, Demetri Terzopoulos  
*Proc. of 25th International Conference on Pattern Recognition (ICPR), Milan, Italy, January 2021, 4465–4472*

## Deep Learning of Neuromuscular and Visuomotor Control of a Biomimetic Simulated Humanoid

Masaki Nakada, Tao Zhou, Honglin Chen, **Arjun Lakshmipathy**, Demetri Terzopoulos

*IEEE Robotics and Automation Letters*, 5(3), July 2020, 3952–3959. (Date of publication: February 10, 2020.)

## Biomimetic Eye Modeling and Deep Neuromuscular Oculomotor Control

Masaki Nakada\*, **Arjun Lakshmipathy\***, Honglin Chen, Tao Zhou, Xin Ling, Demetri Terzopoulos

*ACM Transactions on Graphics*, 38(6), November 2019, 221:1–14. *Proc. of SIGGRAPH Asia, Brisbane, Australia, November 2019*

(\* equal contribution)

## Teaching Experience

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### Introduction to Computer Graphics (TA) | UCLA CS 174A

*Fall 2017, Winter 2018*

Basic principles behind modern two- and three-dimensional computer graphics systems, including complete set of steps that modern graphics pipelines use to create realistic images in real time. How to position and manipulate objects in scene using geometric and camera transformations. How to create final image using perspective and orthographic transformations. Basics of modeling primitives such as polygonal models and implicit and parametric surfaces. Basic ideas behind color spaces, illumination models, shading, and texture mapping.

### Computer Networks: The Physical Layer (TA) | UCLA CS 117

*Spring 2018, Summer 2018, Fall 2018*

Introduction to fundamental computer communication concepts underlying and supporting modern networks, with focus on wireless communications and media access layers of network protocol stack. Systems include wireless LANs (IEEE802.11) and ad hoc wireless and personal area networks (e.g., Bluetooth, ZigBee). Experimental project based on mobile radio-equipped devices (smart phones, tablets, etc.) as sensor platforms for personal applications such as wireless health, positioning, and environment awareness, and experimental laboratory sessions included.

## Technical Skills

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C++ (Point Cloud Library, OpenSceneGraph, Eigen, ImGUI, OpenGL + GLUT), Python (NumPy, Open3D, Tensorflow, PyTorch, SciPy), C, ROS, Blender, AWS (EC2, S3, EKS, ELB, ECS, CloudFront, Cognito, DynamoDB, MediaConvert, Route 53, ACM, SES, SNS), JavaScript (NodeJS / Express, ReactJS, AngularJS), Docker + Kubernetes, NGINX, Jenkins, Java / Spring, Scala / Play, Elasticsearch, Git

## Graduate Coursework

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Hands: Design and Control for Dexterous Manipulation (CMU 16-848)

*Spring 2022*

Discrete Differential Geometry (Audit) (CMU 15-858)

*Spring 2022*

Graduate Artificial Intelligence (CMU 15-780)

*Spring 2022*

Mobile and Pervasive Computing (CMU 15-821)

*Fall 2021*

Graduate Algorithms (CMU 15-750)

*Spring 2021*

Biomechanics and Motor Control (CMU 16-868)

*Fall 2020*

Computational Photography (CMU 15-862)

*Fall 2020*

Problem Solving and Search (UCLA CS 261A)

*Spring 2018*

Special Topics: Shared Economy Systems (UCLA CS 259)

*Spring 2018*

Artificial Life for Computer Graphics and Vision (UCLA CS 275)

*Winter 2018*

Neural Networks and Deep Learning (UCLA EE 239AS)

*Winter 2018*

Embedded Systems (UCLA CS M213A)

*Fall 2017*

## Non-Research Professional Appointments\*

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### Guru Tutor (Part Time)

*March 2021 — Present*

Cambridge Coaching

### Technical Advisor (Part Time)

*September 2020 — December 2020*

NeuralX Incorporated

### Founding Engineer

*March 2019 — August 2020*

NeuralX Incorporated

**Full Stack Software Engineer**  
Rally Health Incorporated

*January 2017 — September 2017*

**Software Engineering Intern**  
Yahoo! Incorporated

*June 2015 — September 2015*

**Software Engineering Intern**  
ViaSat Incorporated

*June 2014 — September 2014*

**iOS Development / Quality Control Intern**  
Sandia National Laboratories

*July 2013 — September 2013*

(\*job titles only - full description and contributions available on LinkedIn)

## Service

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Reviewer: IROS (2021)

CMU CSD Open House Planning Committee (2022)

CMU CSD Introductory Course Planning Committee (2021)

UCLA UPE + TBP Free Tutoring (2016,2015,2014,2013)

## Honors and Awards

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Cum Laude Honors (BS), Upsilon Pi Epsilon Member, Tau Beta Pi Member

## References

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1. **Nancy Pollard**  
Professor  
Computer Science Department, Robotics Institute  
Carnegie Mellon University  
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2. **Demetri Terzopoulos**  
Distinguished Professor and Chancellor's Professor of Computer Science  
Computer Science Department  
University of California, Los Angeles  
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3. **Masaki Nakada**  
Co-Founder and Co-CEO  
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